

```

use IEEE.NUMERIC_STD.ALL;
use work.lava.all;
use work.all;
entity halfAdd is
port(clk : in std_logic;
i0 : in std_logic;
i1 : in std_logic;
carry : out std_logic;
sum : out std_logic);
end entity halfAdd;
architecture str of halfAdd is
signal sig_6_o0 : std_logic;
signal sig_7_o0 : std_logic;
signal sig_2_o0 : std_logic;
signal sig_3_o0 : std_logic;
signal sig_5_o0 : std_logic;
signal sig_4_o0 : std_logic;
begin
sig_6_o0 <= sig_7_o0;
sig_7_o0 <= (sig_4_o0 xor sig_5_o0);
sig_2_o0 <= sig_3_o0;
sig_3_o0 <= (sig_4_o0 and sig_5_o0);
sig_5_o0 <= i1;
sig_4_o0 <= i0;
carry <= sig_2_o0;
sum <= sig_6_o0;
end architecture str;

```

halfAdd_tb.vhd file is listed below:

```

library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_textio.all;
library std;
use std.textio.all;
library work;
entity halfAdd_tb is
begin
end entity halfAdd_tb;
architecture sim of halfAdd_tb is
signal clk : std_logic := '1';
signal rst : std_logic := '0';

```

```

constant input_size : integer := 16;
constant output_size : integer := 16;
signal input : std_logic_vector(3 downto 0) := (others => '0');
signal output : std_logic_vector(3 downto 0);
begin
runtest: process is
FILE halfAdd_input : TEXT open read_
mode IS "halfAdd.in.tbf";
FILE halfAdd_output : TEXT open write_
mode IS "halfAdd.out.tbf";
VARIABLE line_in, line_out : LINE;
variable input_var : std_logic_vector(3
downto 0);
variable output_var : std_logic_
vector(3 downto 0);
variable needs_rst : boolean := false;
begin
while not endfile (halfAdd_input) loop
READLINE(halfAdd_input, line_in);
READ(line_in, input_var);
clk <= '1';
wait for 5 ns;
input <= input_var;
if needs_rst then
rst <= '1';
end if;
output(3 downto 2) <= input_var(3
downto 2);
wait for 20 ns;
clk <= '0';
wait for 20 ns;
if needs_rst then
rst <= '0';
needs_rst := false;
end if;
output_var := output;
WRITE(line_out, output_var);
Writeln(halfAdd_output, line_out);
wait for 5 ns;

```

EFY Note

The source codes of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com

```

end loop;
wait;
end process;

dut: entity work.halfAdd
port map (
clk => clk,
i1 => input(2),
i0 => input(3),
sum => output(0),
carry => output(1)
);

```

end architecture sim;

halfAdd.do file is listed below:

```

vlib mywork
vcom -work mywork Lava.vhd
if [catch {vcom -work mywork halfAdd.
vhd} info] {
puts $info
} else {
vcom -work mywork halfAdd_tb.vhd
vsim -lib mywork halfAdd_tb
add wave -r /*
run -all
}
quit

```

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